TRACK C: CONSUMER, FMCG & RETAIL

Team: **Who cares** | Author: **Abhay Singh & Team**

ARGUS: Autonomous Financial Discovery Agent

The agent finds where reported company growth separates from real business reality, then proves or disproves it.

Deliverables: GitHub Repository & Code • 2-Minute Video Demonstration • Round 1 Submission (ATMOS 2026)

1. IDEA & APPROACH: WHY TRUE DISCOVERIES BEAT BASIC AI SUMMARIES — CORE INSIGHT

Most business AI tools today are just **summarizers**: you paste a 30-page company speech, and the AI repeats whatever good news the CEO said. But company leaders are paid to sound positive. The real business story is almost always hidden in the gap between what executives *say* and what their audited accounts and store numbers *actually show*.

ARGUS is not a chatbot or a summarizer. It operates like an independent investigative auditor through six simple actions: **Discover** (spot unusual gaps), **Connect** (link executive words with official balance sheets), **Calculate** (run exact math with no guesswork), **Reason** (test whether growth is genuine), **Challenge** (play devil's advocate to find fair explanations), and **Prove** (deliver clear findings with clickable public links).

> **Strict Rules:** Zero stock tips or investment recommendations (no "buy", "sell", "hold", or price targets). 100% public information only—nothing paid or behind a password.

2. TOOLS & STACK: REAL TECHNOLOGY (NOT JUST VAGUE "AI") — ARCHITECTURE

Tool / Component	What Powers It	Why It Was Chosen
Public Filings Harvester	Screener.in Financial Tables & BSE India PDF Announcements	100% free, public records of audited quarterly P&L, balance sheet schedules, and investor call transcripts.
Autonomous Agent Framework	Python 3 Pipeline (Urllib, HTML Parsers, RegEx Engine)	Automates source collection, deterministic calculations and evidence-based hypothesis testing, with explicit handling of missing or conflicting data.
Deterministic Math Engine	Standard Python Arithmetic Module (No LLM Math)	Ensures exact calculations for growth rates, DSO, and turnover days without relying on language model math guesswork.
Reasoning & Red-Teaming	Gemini API (Structured Analytical Prompts)	Formulates investigative questions and stress-tests findings against fair industry counter-arguments.
Single Output Interface	Interactive HTML5 Dashboard with Inline SVG Charts	Clean, responsive single-screen dashboard with clickable source links and one-click PDF Executive Memo printing.

3. WORKFLOW: FROM COMPANY NAME TO VERIFIED DISCOVERY — 5-STEP PROCESS

1. Company — Select or enter ticker (e.g. HINDUNILVR, TRENT) → **2. Investigation** — Pick weak spots: Price hikes vs Volume? → **3. Research Plan** — Harvest 12 quarters of P&L, notes & calls → **4. Evidence** — Cross-examine sources: CEO + Math + Rivals → **5. Discovery** — 1-sentence finding + exact disproof rule

How it thinks: The agent does not read aimlessly. It immediately asks: *"Are higher sales coming from selling more goods, or did they just raise prices? Are distributors taking longer to pay their bills? Are new store openings hiding slowing sales at older stores?"*

4. TRACK & GENERALITY: ELIGIBLE TRACK C UNIVERSE — COMPANY-AGNOSTIC RESEARCH

The research pipeline uses company-agnostic analysis rules for eligible NIFTY 50 companies in Track C. It can extend to other consumer-sector peers where permitted and where comparable public data is available:

A. Factory Brands (FMCG Makers)

E.g., Hindustan Unilever, Britannia, Dabur, Nestle, Bikaji (outside NIFTY 50).

Checks if higher revenue is just from raising prices while unit volumes are sluggish, or if warehouse stock is piling up as distributors take longer to clear products.

B. Retail Stores & Jewellery (Non-Volume Disclosers)

E.g., Trent (Westside/Zudio), Titan (Jewellery/Watches), V2 Retail, DMart.

Retailers do not report volume in kilograms. ARGUS tracks whether opening new stores creates an illusion of growth while older stores experience maturing sales throughput.

EVIDENCE, PROOF & VALIDATION

Team: **Who cares**

Cross-Examination, Self-Falsification & The Working Dashboard

5. TRIANGULATION: HOW CLAIMS ARE FACT-CHECKED EVIDENCE TRIANGULATION

Claims are cross-checked against independent public sources, financial filings and peer or industry evidence where available. Missing evidence and unresolved contradictions are explicitly reported:

Leg of Evidence	What We Inspect	How a Contradiction is Caught
Leg 1: Leadership Story	CEO statements in investor presentations and quarterly earnings calls.	The CEO announces: *"Customer demand is accelerating across our retail networks."*
Leg 2: Audited Math	Official balance sheet notes and quarterly cash reports filed with regulators.	Official balance sheet schedules reveal network inventory grew +68.2% (₹2,289 Cr), locking up working capital in unsold inventory.
Leg 3: Industry Rivals	Reports from direct competitors during the same quarter (where available).	Peer disclosures show steady inventory velocity, indicating that inventory stretch is company-specific rather than a macro slowdown.

6. SELF-FALSIFICATION: THE "CHANGE-MY-MIND" RULE COUNTERARGUMENT & FALSIFICATION

Most human analysts suffer from *confirmation bias*—they fall in love with their theory and ignore contrary facts. ARGUS avoids this through two built-in sanity checks:

1. The Fair Alternative (Playing Devil's Advocate)

The agent actively looks for honest business explanations. For example: *"Did inventory rise because management bought raw materials early to protect against upcoming price inflation, or did new stores just need normal opening stock?"*

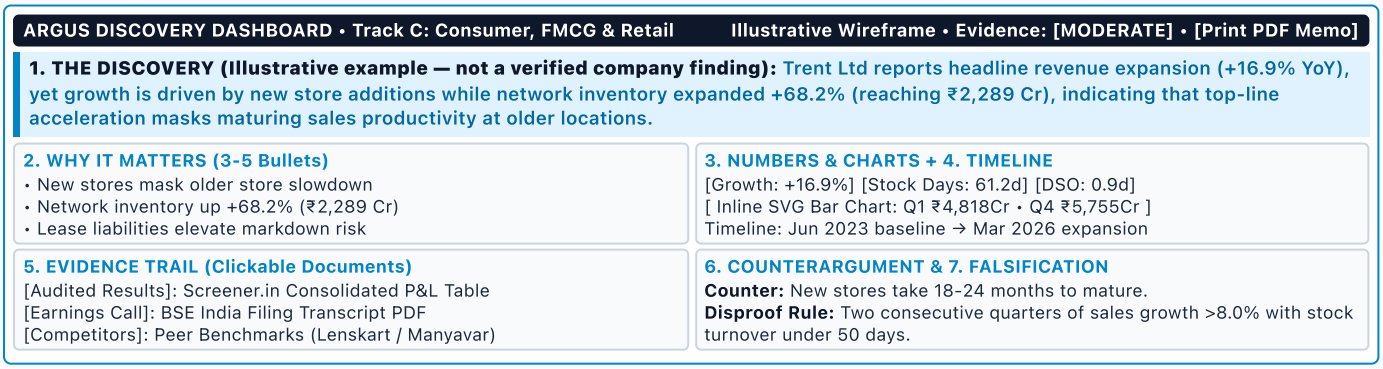
2. The Exact Disproof Condition (Measurable & Time-Bound)

The system sets an exact milestone that would make it admit it was wrong. Example: *"This concern is disproven if the company delivers **two consecutive quarters** of verified volume growth over +4.0% with unpaid bills shrinking."*

7. OUTPUT FORMAT: SINGLE INTERACTIVE DASHBOARD (THE 8 ELEMENTS) INTERACTIVE DASHBOARD

The system delivers a single output: an **Interactive Financial Research Dashboard** with an instant "Export Executive Memo" button. It contains all **eight required elements**:

- **1. The Discovery:** Exactly 1 quantified sentence showing the hidden gap.
- **2. Why It Matters:** 3 to 5 clear bullets on financial sustainability.
- **3. Key Numbers:** Exact calculations and inline SVG visual trend charts.
- **4. Timeline:** When the shift started and how it developed over quarters.
- **5. Evidence Trail:** Clickable links with document title, page, date & excerpt.
- **6. Counterargument:** The strongest alternative business explanation.
- **7. Falsification Condition:** Specific numbers that would disprove the finding.
- **8. Evidence Strength:** Strong, Moderate, or Weak with full explanation.



8. ROADMAP: BUILD PLAN & MILESTONES TO 25 OCT BUILD PLAN & MILESTONES

Date / Milestone	Status: Built vs. Planned	What It Delivers for the Finals
Oct 19, 2026 (Today) **Working MVP**	**BUILT & TESTED** Autonomous execution.	Pulls live filings, runs deterministic math, detects retail/FMCG differences, tests counterarguments, and renders the 8-element dashboard.
Oct 20–21 **Concall Deep Dive**	**IN PROGRESS**	Extracts exact spoken sentences and speaker names from full-length concall PDFs using fast local text matching.
Oct 22–23 **Peer Matrix**	**PLANNED**	Broadens automated cross-checks from 3 peers to 10+ peers across non-NIFTY mid-caps and regional competitors.
Oct 24–25 **Finals Ready**	**PLANNED**	Stress-tested on 50+ Indian consumer tickers to deliver a flawless live 10-minute presentation on whichever company the judges pick.